

The Worked-Systems Audit: Exact Counts, Dual Witnesses, and Design Rules for the Obstruction Programme’s Catalogue

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Abstract

This paper is the programme’s worked-systems flagship: one audit frame under which every machine-verified count, witness, and design rule of the catalogue is stated once, with one notation and one verification chain. It consolidates the selector-principle concordance (the shared audit system, the corrected delay identity with its boundary case, the coarsest-partition refutation, the four-cell and decentralized and mode-blind concordances, the certainty-equivalence convention, and the continuous benchmark), the static duality edition (the two-floor instance, the normalized Farkas pair, the convexity gap, the Helly-tight triple, and the benchmark caps), the monitoring design rules with the non-strict review-timing identity, and the four audited instance lines (vector floors, the policy-class lattice, discrete regimes, and decentralized observation). No new theorem is claimed: every statement is a pointer to a source edition whose script verifies it, and one master script chains all seven lineage scripts (57 checks) with its own checks on the text.

Keywords: viability kernels; obstruction certificates; exact arithmetic; worked systems; monitoring design; verification discipline

1. Introduction

The obstruction calculus (*An Obstruction Calculus for Viability under Incomplete Observation*, the programme’s theory flagship) states its certificates mechanism by mechanism and consolidates them once as dual witnesses of the recursively viable selector. The programme’s remaining published output has been organized the other way around: one short work per instance line, each with its own notation, its own script, and its own addendum. That organization is verification-grade but scattered, and the programme’s consolidation decision (the follow-up roadmap, v17) assigns the catalogue facts a single home. This paper is that home.

The claim type of this flagship is the exact audit. What is consolidated here is not new theory but the audited numbers themselves: for each worked system, the state space, the threshold convention, the exact count or identity, the witness, and the script that regenerates it. Where a source edition corrected an earlier claim, the corrected

form is the one stated, and the correction is part of the audited record — the strict-form delay rule is stated only to record where it fails; the “partition into maximal common-action sets” is stated only as a refuted reading; the certainty-equivalence drift labels carry their round-four corrections.

2. The audit frame

Three conventions govern everything that follows.

Exact arithmetic. Every count below is computed in integer or rational arithmetic by a deterministic, standard-library script; no solver, no floating-point tolerance, no simulation. The scripts are named per section and chained by the master verification (Section 8).

The shared audit system. A large fraction of the catalogue instantiates one three-state-coordinate system (Abaee, 2026a, Section 8): states $x = (z_1, z_2) \in \{0, 1, 2, 3\}^2$, safe set $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$, instruments $u \in \{0, 1, 2\}$ with transition $z^+ = \min(3, z + 1)$ on the acted coordinate’s regeneration and $\max(0, z - 1)$ elsewhere (the unacted coordinate loses one; $u = 0$ executes no instrument and both coordinates lose one). The corner $(1, 1)$ is the physically nonviable state, and the nine pair states of $\mathcal{V}$ generate $C(9, 2) = 36$ audited belief pairs at review horizon $H = 12$. Observation structures (partitions, laws, regimes) are variations on this system; the continuous benchmark of Section 4 is its polyhedral cousin.

Threshold and drift conventions. The review-timing threshold is $\sigma^*(z_0) = z_0 - 1$ on the hold class, and the monitoring identity is non-strict (Section 5). The canonical *uncorrected* certainty-equivalence law is the one whose drift is bounded below by $21/100$ on $[1, 2]$; the corrected law’s drift is identically zero (Section 3).

3. The concordance (the shared system, audited)

Source edition: *The Selector Principle and Its Dual Witnesses: A Concordance for the Obstruction Programme v1* ([paper2_selector_concordance_v1](#), script 8/8). Its eight checks, restated once:

C1 (the delay identity and its boundary). On the 48-cell audit grid (z_0, T_{obs}) , viability at the review deadline holds if and only if $T_{\text{obs}} \leq \sigma^*(z_0) = z_0 - 1$: zero mismatches.

The strict form $T_{\text{obs}} < \sigma^*(z_0)$ fails at exactly the boundary cell (2, 1), which is viable; the boundary-inclusive form is the corrected rule.

C2 (adequacy and the refuted coarseness rule). Over all 15 partitions of the target, fibre-admissibility holds exactly when singleton admissibility does; two minimal two-cell adequate partitions exist; and the two maximal common-action sets $\{(1, 2), (2, 2)\}$ and $\{(2, 1), (2, 2), (3, 1)\}$ overlap in (2, 2), so a monitoring design “partitioning into maximal sets” does not exist. The coarsest-adequate-monitoring rule of the earlier monitoring edition is corrected accordingly.

C3 (the four-cell policy-class concordance). With the no-repeat codex threaded as a parameter through the winning recursion: institutional (aggregate, codex) 24 of 36 pairs; aggregate per-cell 26; full-information codex 25; full information 28. The meet of the two middle cells is the institutional kernel setwise; the corner (1, 1) is nonviable as a singleton, and the top kernel is exactly the $C(8, 2)$ pairs avoiding it.

C4 (decentralized observation). Unanimity over the 256-law space ($2^4 \times 2^4$: four coordinate values, two votes per agency) sustains 12 of 36 pairs; exactly 16 coordinated-viable pairs are lost; 16 of the 256 law pairs sustain the audited belief — a coincidence of counts with the loss count, not an identity; and forbidding agency 2’s vote collapses the kernel from 12 to 0.

C5 (discrete regimes). With frozen regimes the corner is viable under each, and the frozen kernel is $36 = 36$; the mode-blind kernel is exactly 28 and coincides setwise with the base full-information kernel under a different damage routing — recorded as a coincidence.

C6 (the certainty-equivalence convention). The canonical uncorrected law’s drift is at least 21/100 on a 101-point grid of $[1, 2]$; the corrected law’s drift is identically zero.

C7 (the threshold convention). The hold-class discrete survival supremum is $\lfloor z_0 - 1 \rfloor$ on the grid, with level sets exactly $\{z_0 \geq 1 + k\}$.

C8 (the continuous benchmark). The critical aggregate is $Y^* = 27/5$ (the cap sum equals 2 exactly there); the $Y = 5$ fibre is viable with witness (6/5, 4/5); the $Y = 6$ fibre is obstructed with cap sum $47/25 < 2$ and dual measure (1/2, 1/2) on the two active-floor atoms, margin 3/50.

4. Static duality and sparse witnesses

Source edition: *Minimax Dual Certificates* v1 ([minimax_dual_certificates_v1](#), script 8/8). The audited static core: on the two-floor instance the max–min equals the min– λ max at $-1/10$; the multiplier $\lambda = (1/2, 1/2)$ is the normalized Farkas pair of the calculus’s worked certificate; the convexity gap instance $U = \{0, 1\}$ with rows $(-1, +1)$, $(+1, -1)$ has an empty common set, max–min -1 against min– μ max 0; the Helly-tight triple $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ exhibits the $m+1$ -point sparse witness; and the benchmark caps $\text{cap}_1 = 3/2 - (Y - 2)/10$, $\text{cap}_2 = 59/50 - (Y - 2)/10$ give viability exactly at $Y \leq 27/5$, with fibre width $\Delta x_1(Y) = Y - 4$. Weak duality throughout; strong duality exactly in the finite and static classes.

5. Monitoring design rules

Source edition: *Monitoring Design* v3 ([monitoring_design_v3](#), script 9/9), the round-six repair edition. The audited rules, as identities on the shared system:

- *Timing (E4, E9).* Viability at the deadline holds exactly when $T_{\text{obs}} \leq \sigma^*(z_0)$, boundary included; the strict form fails at exactly one grid cell (C1’s cell), and the repair is pinned by check E9.
- *Coarseness (E5).* A fibre merging states with disjoint safe action sets obstructs at that information state while every branch remains viable; the remedy is a fibre split, not a finer threshold on the same index.
- *Aggregation (E6, C3).* An index whose fibres cross a floor cannot support an exact safe/unsafe verdict; the certainly-safe set is the honest report.
- *Bias (E7, C6).* The uncorrected law empties a nonempty kernel under an injective observation map; the remedy is structural correction of the observation map, not a better sensor.

The scoping check E8 records that the design rules are stated on the audited systems; E1–E3 pin the grid, the fibre intersections, and the codex.

6. The instance catalogue

Four instance lines, each consolidated from its second (audited) edition.

6.1 Vector floors ([vector_floor_certificates_v2](#), script 8/8). Under multiple sustainability floors the Farkas certificate obstructs jointly: the multi-constraint obstruction is a single certificate over all floors, not one constraint at a time (the first edition’s abstract claimed otherwise and was corrected). The minimal infeasible subsystem reading (Remark 1) carries witnesses; E7 verifies the pair family with the corner (1, 1)’s dynamic nonviability; E8 pins the corner fact. The line’s first-breach timing bound and corner-cone reading are recorded as successor-layer objects.

6.2 The policy-class lattice ([policy_class_lattice_v2](#), script 8/8). The four audited policy classes form the Boolean lattice $\{0, 1\}^2$ with incomparable middle cells: the audit’s missing fourth cell (full information under the no-repeat codex) holds exactly 25 of 36 pairs (witnesses $\{(1, 3), (3, 1)\}$ and $\{(1, 3), (2, 1)\}$), against the aggregate’s 26 (witnesses $\{(1, 2), (3, 1)\}$, $\{(1, 3), (2, 1)\}$). The kernel meet equals the institutional kernel setwise and the kernel join is the full-information kernel (28, C3’s top). The corrected certainty-equivalence labelling (21/100 uncorrected, zero corrected) is programme-wide from this edition.

6.3 Discrete regimes ([hybrid_mode_viability_v2](#), script 8/8). Mode uncertainty loses value only through singletons: the corner (1, 1) is the only mode-blind-dead state, every lost pair contains it, and no merged-pair loss exists (E7 verifies all nine singleton verdicts). The causal timing is fixed (the mode realized at step t is the adversary’s end-of-step- $(t - 1)$ choice), and the mode-blind kernel’s setwise coincidence with the base kernel (C5) is audited as a coincidence under different damage routings, with the corner’s frozen-regime viability pinned.

6.4 Decentralized observation ([institutional_observation_v2](#), script 8/8). Votes on a shared instru-

ment: the law space is $256 = 2^4 \times 2^4$, the unanimity kernel 12 of 36, the loss instance $\{(1, 2), (3, 1)\}$ pinned, the deadlock transition ($u = 0$: both coordinates lose one) declared, agency 2’s vote on its own previous coordinate is a delayed own reading (local memory, not common knowledge), and forbidding that vote collapses the kernel $12 \rightarrow 0$ (C4). Decentralizability is an existence certificate over the law space.

7. Absorbed lines and the practitioner reading

The remaining catalogue lines are absorbed by pointer with their editions retained: finite-horizon completeness, certificate duality, comparison drift–exit, buffered (approximate) viability, and hidden-parameter learning — each with its script in the chain below — together with the viable-selector and five-layer successor lineages (absorbed into the theory flagship’s Sections 7–8), the stochastic-selector line (the probabilistic flagship’s seed), and the applications note with its utility record, whose limitation register (verification-grade software; no calibrated case; completeness scope; single-system scope) this flagship’s provenance tables now partly discharge. For the practitioner: the catalogue’s operational reading is the four monitoring rules of Section 5 plus the threshold convention C7 — the review must land no later than worst-case exit, boundary included; a merged fibre is diagnosed by a split, not a threshold; aggregate indices report the certainly-safe set or fall silent; and a biased observation map is repaired structurally.

Source edition (retained)	Audited here
selector concordance v1	C1–C8 (§3)
minimax dual certificates v1	duality core (§4)
monitoring design v3	rules E1–E9 (§5)
vector floors v2	§6.1
policy-class lattice v2	§6.2
hybrid modes v2	§6.3
institutional v2	§6.4
five absorbed lines (§7)	chained scripts (§8)

Table 1: The consolidation registry. Every source edition is retained unmodified in the repository; supersession is recorded, never deletion.

8. Verification discipline

One master script, `paper2_worked_systems_v1_verification.py` (standard library only), does two things. First, it chains the seven lineage scripts listed in Table 1 as subprocesses and asserts each exits green: $8 + 8 + 9 + 8 + 8 + 8 + 8 = 57$ checks. Second, it runs its own checks: the headline counts of Sections 3–6 are recomputed from the shared system’s primitives (the pair family, the codex-threaded winning recursion with the no-repeat rule threaded as a parameter, the law space, and the benchmark caps), and every number printed in this text is asserted present. A claim in this paper is admissible exactly when both layers pass.

9. Conclusion

The catalogue now has one home. The consolidation changes no verdict: the corrected delay identity stands with its boundary case; the coarsest-partition rule stands refuted; the four-cell lattice stands with its incomparable middles; the institutional and mode-blind kernels stand with their audited coincidences; and the design rules stand as identities on the audited systems. What changes is the address: one audit frame, one notation, one verification chain, and a registry that records every source edition as retained. The theory flagship carries the principles; this flagship carries the numbers; and the remaining flagships (computational, probabilistic) seed from the lineages already verified here.

References

Abae, A. Robust viability of the 2J3KL limit reference point under a surplus-production map. Zenodo. <https://doi.org/10.5281/zenodo.22552060> (the shared audit system’s source, Section 8).

Programme editions consolidated: `paper2_selector_concordance_v1`; `minimax_dual_certificates_v1`; `monitoring_design_v3`; `vector_floor_certificates_v2`; `policy_class_lattice_v2`; `hybrid_mode_viability_v2`; `institutional_observation_v2`; and the absorbed lines of Section 7 — all publicly archived in the repository <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`).

Declarations

Funding. None. Competing interests. None. Data availability. No external data were used. Code availability. The verification script `paper2_worked_systems_v1_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it chains the seven lineage scripts and reproduces all consolidated checks verbatim. AI declaration. GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review, under the programme’s verification discipline.